

CUI

PORTSMOUTH NAVAL SHIPYARD



FY 25 DEPOT MAINTENANCE AVAILABILITY (DMP-1)

UPGRADE AND REPAIR THE SECONDARY PROPULSION UNIT (SPU) ROTATIONAL
DEPLOYMENT MECHANISM (RDM), ROTARY JOINT AND HYDRUALIC TRAINING MOTOR

SPECIFICATION FOR SERVICE AND WORK TO BE ACCOMPLISHED

CONTROLLED BY: Portsmouth Naval Shipyard
CONTROLLED BY: Code 220P
CUI CATEGORY: OPSEC
DISSEMINATION CONTROLS: FEDCON



CUI

ITEMS THAT ARE STRUCK OUT IN THIS INDEX DO NOT APPLY TO THIS CONTRACT.

CATEGORY I FY-24 STANDARD ITEMS APPLICABLE TO THIS JOB ORDER WITHOUT FURTHER REFERENCE.

<u>ITEM NO.</u>	<u>TITLE</u>	<u>DATE</u>
009-01	General Criteria; accomplish	25OCT2022
009-02	Environmental Compliance Report for Material Usage; accomplish	25OCT2022
009-03	Toxic and Hazardous Substance; control	25OCT2022
009-04	Quality Management System; provide	25OCT2022
009-05	Temporary Access; accomplish	25OCT2022
009-06	Maintaining Protection and Cleanliness from Non-Radioactive Operations; accomplish	25OCT2022
009-07	Confined Space Entry, Certification, Fire Prevention and Housekeeping; accomplish	25OCT2022
009-08	Shipboard Fire Protection and Fire Prevention; accomplish	25OCT2022
009-09	Process Control Procedure (PCP); provide and accomplish	27FEB2022
009-10	Asbestos – Containing Material (ACM); control	25OCT2022
009-11	Insulation and Lagging; accomplish	25OCT2022
009-12	Weld, Fabrication, and Inspect; accomplish	25OCT2022
009-13	Meter, Gage, Switch and Thermometer; repair	25OCT2022
009-15	Rotating Machinery; balance	25OCT2022
009-16	Electronic Equipment; repair	25OCT2022
009-17	Rotating Electrical Equipment; repair	25OCT2022
009-18	Mine Warfare Ships Magnetic Material; control	25OCT2022
009-19	Provisioning Technical Documentation (PTD); provide	25OCT2022
009-20	Government Property; control	25OCT2022
009-21	Alteration Verification, Logistics and Technical Data; provide	25OCT2022
009-23	Interferences; remove and install	25OCT2022
009-24	Authorization, Control, Isolation, Blanking, Tagging and Cleanliness; accomplish	25OCT2022
009-25	Structural Boundary Test; accomplish	25OCT2022
009-26	Deck Covering; accomplish	25OCT2022
009-27	Material Identification and Control (MIC) for Level I System; accomplish	25OCT2022
009-30	Boiler Sample Tube; inspect	25OCT2022
009-31	Boiler Waterjet Cleaning; accomplish	25OCT2022
009-32	Cleaning and Painting Requirements; accomplish	25OCT2022
009-33	Rotating Electrical Equipment; rewind	25OCT2022
009-34	Fire Protection of Unmanned Vessels at Contractor's Facility; accomplish	25OCT2022
009-36	Controller; repair	25OCT2022
009-37	General Procedure for Woodwork; accomplish	25OCT2022
009-38	Boiler, Catapult Accumulator, and Reboiler Dry Lay up; accomplish	25OCT2022
009-39	Technical Manual Contract Requirement (TMCR) for New Technical Manuals for Commercial Equipment/Component; accomplish	25OCT2022
009-40	Contractor Crane, Multi-Purpose Machine and Material Handling Equipment at a Naval Facility; provide	25OCT2022
009-45	Tapered Plug Valve; repair	25OCT2022

[REDACTED]		
<u>ITEM NO.</u>	<u>TITLE</u>	<u>DATE</u>
009-46	Butterfly Valve, Synthetic and Metal Seated; repair	25OCT2022
009-47	Gate Valve; repair	25OCT2022
009-48	Pressure Seal Bonnet Valve Shop Repair; accomplish	25OCT2022
009-49	Pressure Seal Bonnet Valve In-line Repair; accomplish	25OCT2022
009-50	Horizontal Swing Check Valve; repair	25OCT2022
009-52	Relief Valve; repair	25OCT2022
009-53	Bolted Bonnet, Globe, Globe Angle, and Globe Stop Check Valve Shop Repair; accomplish	25OCT2022
009-54	Bolted Bonnet, Globe, Globe Angle, and Globe Stop Check Valve In-line Repair; accomplish	25OCT2022
009-55	Regulating/Reducing Valve; repair	25OCT2022
009-56	Main Propulsion Boiler Wet Lay-up; accomplish	25OCT2022
009-57	Reduction Gear Security; accomplish	25OCT2022
009-58	Pump and Driver Shaft Alignment; accomplish	25OCT2022
009-60	Schedule and Associated Reports for Availabilities over 9 Weeks in Duration; provide and manage	25OCT2022
009-61	Shipboard Use of Fluorocarbons; control	25OCT2022
009-62	Boiler Handhole, Manhole Seat and Plate; inspect	25OCT2022
009-63	Lubricating Oil and Hydraulic Fluid; analyze	25OCT2022
009-65	Polychlorinated Biphenyls (PCBs); control	25OCT2022
009-67	Integrated Total Ship Testing; manage	25OCT2022
009-69	Heavy Weather/Mooring Plan; provide	25OCT2022
009-70	Confined Space Entry, Certification, Fire Protection, Fire Prevention and Housekeeping for Unmanned Vessels; accomplish	25OCT2022
009-71	Piping System; test	25OCT2022
009-72	Physical Security at Private Contractor's Facility; accomplish	25OCT2022
009-73	Shipboard Electrical/Electronic Cable Procedure; accomplish	25OCT2022
009-74	Occupational, Safety and Health Plan; accomplish	25OCT2022
009-75	Circuit Breaker; repair	25OCT2022
009-76	Waveguide and Rigid Coaxial Lay Up; accomplish	25OCT2022
009-77	Cofferdam Installation; accomplish	25OCT2022
009-78	Passive Countermeasures System (PCMS) Material; repair	25OCT2022
009-79	Government Owned Material (GOM); Status for Multi Ship Multi Option availabilities; report	25OCT2022
009-80	Ship's Facilities; maintain	25OCT2022
009-81	Compartment Closeout; accomplish	25OCT2022
009-82	Installation of Equal Component Vice Specified Component; report	25OCT2022
009-83	Wire Rope Assembly; fabricate	25OCT2022
009-84	Threaded Fastener Requirements; accomplish	25OCT2022
009-85	Government Sponsored Planning Yard/Configuration Data Manager (CDM) On-Site Representative Facility; provide	25OCT2022
009-86	Recovery and Turn-In of Ozone Depleting Substance (ODS); accomplish	25OCT2022
009-87	Chemical Disinfection Procedure; accomplish	25OCT2022
ITEM NO.	TITLE	DATE

009-88	Collection, Holding and Transfer (CHT) and Motor Gasoline (MOGAS) Tanks, Spaces, and Piping, including Sewage or MOGAS Contaminated Tanks, Spaces, and Piping; certify	25OCT2022
009-90	Technical Representative; provide	25OCT2022
009-91	Propeller In-Place Inspection; accomplish	25OCT2022
009-92	Resilient Mount; install	25OCT2022
009-93	Emergency Planning and Community Right to Know Act (EPCRA) and Pollution Prevention Act (PPA) Information; provide	25OCT2022
009-95	Mechanically Attached Fitting (MAFs); install	25OCT2022
009-96	Ball Valve; repair	25OCT2022
009-99	Ship Departure Report; provide	25OCT2022
009-100	Ship's Stability; maintain	25OCT2022
009-101	Ship Transit and Berthing; accomplish	25OCT2022
009-103	Weight and Moment Change Data; provide	25OCT2022
009-104	Vibration Testing and Analysis; accomplish	25OCT2022
009-105	Thermal Sprayed Coating for Machinery Component Repair; accomplish	25OCT2022
009-106	Work Authorization Form Coordinator (WAFCOR); provide	25OCT2022
009-107	Piping System Cleanliness Restoration and Flushing (Non-Nuclear); accomplish	25OCT2022
009-109	Non-SUBSAFE Work on SUBSAFE-Certified Vessel; accomplish	25OCT2022
009-110	Non-Nuclear Work on a Nuclear Vessel; accomplish	25OCT2022
009-111	Scheduled and Associated Reports for Availabilities 9 Weeks or Less in Duration; provide and manage	25OCT2022
009-112	Prevention of Radiographic Inspection Ionizing Radiation Hazard; accomplish	25OCT2022
009-113	Rotating Electrical Equipment with a Sealed Insulation System (SIS); rewind	25OCT2022
009-114	Mold Remediation; accomplish	25OCT2022
009-115	Bearing Re-babbiting; accomplish	25OCT2022
009-116	Waste Heat Boiler Sodium Nitrite Wet Lay-up; accomplish	25OCT2022
009-117	Combat Systems Light Off Support; provide	25OCT2022
009-118	CG Deck Loading; accomplish	25OCT2022
009-120	Fact Finding and Critique of Unplanned Events; manage	25OCT2022
009-121	Ship Assessment / Inspection; accomplish	25OCT2022
009-122	Temporary Pad-eye; install and remove	25OCT2022
009-123	Fiber Optic Component; remove, relocate, repair, and install	25OCT2022
009-124	Thermal Spray Non-Skid Application; accomplish	25OCT2022
009-125	Boats Less Than 65 Feet Long; accomplish	25OCT2022

CUI

ITEM NO.

TITLE

238-02-001

Secondary Propulsion Unit (SPU) Rotatable Deployment Mechanism (RDM)

1. SCOPE:

- 1.1. Title: Upgrade and repair the Secondary Propulsion Unit (SPU) Rotary Deployment Mechanism (RDM).
- 1.2. Location of Work:
 - 1.2.1. Curtiss Wright (CW) Electro-Mechanical Division (EMD) repair facility.
- 1.3. Identification:
 - 1.3.1. Quantity: (1 EA), Rotational Deployment Mechanism (RDM) and support material including but not limited to the following items:
 - 1.3.1.1. Rotary Joint
 - 1.3.1.2. Hydraulic Training Motor

2. REFERENCES:

- 2.1. Standard Items, (FY-24)
- 2.2. Submarine Maintenance Standard (SMS) 2380-081-029 Rev C, Chg. 1, Restore Secondary Propulsion Rotational Deployment Mechanism.
- 2.3. SMS 2380-081-027 Rev –, Chg. 5, Restore Secondary Propulsion System Rotational Deployment Mechanism Rotary Joint.
- 2.4. DFS [REDACTED] 1056-2018, [REDACTED] RDM TRAINING MOTOR BREAKAWAY TORQUE VIOLATES REQMTS. PERM DFS
- 2.5. LAR SHAPEC-0026-25, [REDACTED] CLASS ROTARY JOINT CONFIGURATION APPLICABILITY
- 2.6. Curtiss-Wright DWG 4D98026 REV 7 / RDM Hydraulic Motor

3. REQUIREMENTS:

- 3.1. Curtiss Wright- EMD (CW) shall provide labor, management, supervision, tools, materials, transportation, and equipment necessary to restore/upgrade components as detailed in paragraphs 3.1.1. through and including 3.1.3. below and repair all identified deficiencies as approved by Portsmouth Naval Shipyard (PNSY).
 - 3.1.1. **Rotational Deployment Mechanism (RDM)**
 - 3.1.1.1. CW shall refurbish and (as applicable) upgrade the RDM per Reference 2.2 Enclosure 5 and RDM Refurbishment Repair Procedure RP10748 guidance. Upon completion of work, CW shall ship the RDM to PNSY.
 - 3.1.1.1.1. CW shall provide objective quality evidence that certifies the 24/EA training motor mounting studs (P/N 4541B76H01 of Enclosure 1) and 24/EA training motor mounting nuts (P/N 4541B77H01 of Enclosure 1) have been replaced and assembled in accordance with NOTE 24, NOTE 25 and Paragraph 5.I.(16) of Attachment 5 of Reference 2.2. **(CDRL A001)**
 - 3.1.1.2. Reference 2.2, Attachment 5, Paragraph 6 specifies the in-shop testing that is required for the RDM assembly (including training motor and rotary joint) based on the work accomplished during restoration. CW shall perform testing of the RDM assembly in accordance with paragraph 6 of Reference 2.2.

3.1.1.2.1. For each shop test performed, CW shall provide objective quality evidence including, but not limited to, test medium, test medium temperature (if applicable), test pressure, test duration and descriptive results of the test. (CDRL A002)

3.1.2. Rotary Joint

3.1.2.1. CW shall refurbish the Rotary Joint in accordance with Reference 2.3. Upon completion of work, CW shall ship the Rotary Joint to PNSY.

3.1.2.1.1. PNSY notes that there were reports of significant seawater contamination within the hydraulic fluid system cycling through the rotary joint assembly. During initial disassembly, CW shall document condition of the internal rotary joint assembly components that indicate unsatisfactory conditions due to seawater corrosion, accelerated wear/degradation due to sea growth, contamination etc. Documentation shall include pictures (where necessary), components replaced and specific non-conforming conditions that resulted in replacement. (CDRL A002)

3.1.2.1.2. PNSY notes that Per Reference 2.5, the [REDACTED] Rotary joint assembly was built to revision 8 of DWG 4D98436. If refurbishment work performed by CW results in a final configuration equivalent to newer Revisions of Curtiss Wright Drawings specified in Reference 2.5, CW shall document this in final report to PNSY. Documentation may be shipped separate from the unit, but must be sent within 30 days of component shipping dates via trackable carrier. Documentation may be combined with other documents shipped per this Statement Of Work (SOW).

3.1.2.1.3. CW shall perform testing of the rotary joint in accordance with paragraph 1.m of Reference 2.3. For each shop test performed, CW shall provide objective quality evidence including, but not limited to, test medium, test medium temperature (if applicable), test pressure, test duration and descriptive results of the test. (CDRL A002)

3.1.3. Training Motor

3.1.3.1. CW shall upgrade and repair the Training Motor in accordance with Reference 2.2 and Refurbishment Repair Procedure RP10748. Upon completion of work, CW shall ship the Training Motor to PNSY.

3.1.3.2. PNSY notes that there were reports of significant seawater contamination within the hydraulic fluid system cycling through the training motor assembly. During initial disassembly and inspections, CW shall document condition of the internal training motor assembly components that indicate unsatisfactory conditions due to seawater corrosion, accelerated wear/degradation due to sea growth, hydraulic fluid contamination etc. Documentation shall include pictures (where necessary), components replaced and specific non-conforming conditions that resulted in replacement. (CDRL A002)

3.1.3.2.1. CW shall provide documentation in accordance with paragraph 5.m of Attachment 5 of Reference 2.2. Upon completion, CW shall mail completed documentation to SUBMEPP:

Submarine Maintenance Engineering Planning and Procurement
(SUBMEPP) Activity
PO Box 2500
Portsmouth Naval Shipyard
Portsmouth, NH 03804-2500
ATTN Code 1843AG

3.1.3.2.2. The SSN 779 RDM (S/N 318) hydraulic training motor was previously restored/upgraded by Original Equipment Manufacturer (OEM) vendor CW during EDSRA-1. As documented in Reference 2.4, following Refurbishment, the Hydraulic training motor worm shaft breakaway torque did not meet the requirements of Reference 2.6 and was reported as 36 FT-LBS (6 FT-LBS beyond the maximum allowed). Upon completion

[REDACTED]
of restoration and satisfactory testing, CW shall document the as-released hydraulic motor worm shaft breakaway torque and provide to PNSY. **(CDRL A002)**

- 3.1.3.2.3. CW shall perform testing of the training motor in accordance with Reference 2.2, Attachment 5, Paragraph 1.e and required testing detailed in Refurbishment Repair Procedure RP10748. For each shop test performed, CW shall provide objective quality evidence including, but not limited to, test medium, test medium temperature (if applicable), test pressure, test duration and descriptive results of the test. **(CDRL A002)**

3.1.4. **Materials:**

- 3.1.4.1. CW shall procure material identified in Enclosure 1 for use at CW-EMD facilities during upgrade and refurbishment of the RDM, Training Motor and Rotary Joint.
- 3.1.4.2. CW shall provide objective quality evidence of all new material procured and utilized during upgrade and refurbishment of the RDM, Training Motor and Rotary Joint. **(CDRL A003)**

4. **NOTES:**

- 4.1. Known Points Of Contact for sole source CW:

Dustin Koedel
Curtiss Wright Electro-Mechanical Division
1000 Wright Way
Cheswick, PA 15024
Phone: (724) 275-5000

- 4.2. The material listed in Enclosure 1 that is to be procured per paragraph 3.1.4 includes items that have been identified as long lead time material that the OEM may utilize during refurbishment.

5. **GOVERNMENT RESPONSIBILITIES:**

- 5.1. PNSY shall provide the following services in support of this contract:
- 5.1.1. RDM assembly will be crated and shipped by PNSY in a reusable solid container. RDM assembly shall be shipped back to PNSY utilizing the same container.
- 5.1.2. Additional material that is required per Reference 2.2 during shipboard reinstallation of the RDM by PNSY and is considered long lead time material (i.e. breechlock shock mounts) will be procured by PNSY through the National Stock System.
- 5.2. PNSY as the Lead Maintenance Authority will provide final evaluation and disposition (approval or disapproval) of identified non-conformances.
- 5.2.1. PNSY will provide engineering services as the Lead Technical Authority for this availability which includes, but is not limited to; review of Departures From Specification (DFS), Liaison Action Request (LAR) with VIRGINIA Class Planning Yard (VAPY), communications with NAVSEA Technical Authority (NAVSEA 05), communication with Naval Reactors Representative Office (NRRO, NAVSEA 08), and communications with Reactor Plant Planning Yard (RPPY).

6. **GOVERNMENT FURNISHED MATERIAL (GFM):**

- 6.1. (One EA), Rotational Deployment Mechanism (RDM) (including the Training Motor and Rotary Joint), Part number 4D98024G01.

7. PERIOD OF PERFORMANCE:

- 7.1. The Period of Performance (POP) for this contract is 01MAY 2026 through 01MAY 2027.
- 7.2. RDM assembly and associated items will be provided to CW no later than RM00 for [REDACTED] DMP.
- 7.3. RDM assembly, associated items and any excess new material purchased per this SOW not utilized during refurbishment will be returned to PNSY no later than 6 months prior to UD48 for [REDACTED].
- 7.4. Availability Key Event (KE) and Milestone (MS) dates are not yet established. Final schedule shall be determined at the time of award but shall adhere to all delivery guidelines identified in paragraphs 7.1 through 7.3.

8. DELIVERABLES:

- 8.1. Upon completion of the RDM, Rotary joint and Training motor, CW shall ship completed documentation with the RDM to PNSY. Documentation may be shipped separate from the unit, but must be sent within 30 days of component shipping dates via trackable carrier. Documentation may be combined with other documents shipped per this SOW as requested in the following sections:
 - 8.1.1. 3.1.1.1.1.
 - 8.1.2. 3.1.1.2.1.
 - 8.1.3. 3.1.2.1.1.
 - 8.1.4. 3.1.2.1.3.
 - 8.1.5. 3.1.3.2.
 - 8.1.6. 3.1.3.2.2.
 - 8.1.7. 3.1.3.2.3.
 - 8.1.8. 3.1.4.2.
- 8.2. Upon completion of the RDM, Rotary joint and Training motor, CW shall mail completed documentation to SUBMEPP per 3.1.3.2.1. Documentation may be shipped separate from the unit, but must be sent within 30 days of component shipping dates via trackable carrier.

LONG LEAD TIME MATERIAL PARTS LIST

ROTATIONAL DEPLOYMENT MECHANISM (RDM) MATERIAL			
OEM PART NUMBER	ITEM DESCRIPTION	QTY	UNIT
8575C32H01	THRUST BEARING RING	3	EACH
8575C87H02	INDICATOR THREAD BODY	1	EACH
8575C88H02	INDICATOR THREAD ROD	1	EACH
4542B13H01	GEAR HOUSING LIP SEAL	1	EACH
4542B12H01	INDICATOR LIP SEAL	2	EACH
8576C08H02	INNER GUIDE SLEEVE	2	EACH
8576C45H01	OUTBOARD BRG SPACER CSTG	1	EACH
8577C32H01	RDM MOTOR SHIM MATERIAL	1	EACH
4397B61H03	SLF LOCK BOLT -.50-13X1.5	1	EACH
4541B04H01	INDICATOR NUT	2	EACH
4541B08H02	BOLT -.375-16X1.00 SLFLKG	10	EACH
4541B08H04	BOLT -.375-16X1.38 SLFLKG	10	EACH
4541B37H06	.375-16X2.00 LG. SLFLKG CAPSCREW	6	EACH
4541B77H01	HEX NUT	30	EACH
4542B10H03	CAP SCREW - SLFLKG SOC HD	4	EACH
4542B11H01	HEX NUT - SLFLKG - .75-10	30	EACH
4542B20H02	CAP SCREW 1.0-8 X 2.25LG	10	EACH
4541B75H01	HEX NUT	2	EACH
4D96344G01	AXIAL DEPLOY SHOCK MOUNT	2	EACH
8577C07G01	GEAR SHOCK MOUNT MLD ASSY	16	EACH
4538B51H02	PIN- THR SLV RAD DEPL BRG	2	EACH
4540B93H02	SLIP WASHER	1	EACH
4541B09H02	INDICATOR KEY MACHINING	2	EACH
4541B17H02	GRIP WASHER	1	EACH
4542B34H01	CABLE CLAMP	4	EACH
4D98024H57	PARKER PLUG 8P5N-M	10	EACH
4541B76H01	TRAINING MOTOR MOUNTING STUDS	24	EACH
8579C96H02	ANTIROTATION PIN	1	EACH
4D98024G02	SHOCK MOUNT/BEARING SLEEVE ASSY	2	AY
8576C58H01	SPHERICAL INSERT	1	EACH
4397B60H01	THRUST WASHER	1	EACH
4541B05H02	LOCK PIN	1	EACH
4542B10H05	CAP SCREW - SLFLKG SOC HD	2	EACH

HYDRAULIC TRAINING MOTOR MATERIAL			
OEM PART NUMBER	ITEM DESCRIPTION	QTY	UNIT
P04900700	SEALING SPACER	1	EACH
P04901000	COATED SLEEVE	1	EACH
P05023100	RETAINING RING	1	EACH
P06555500	SPL DRIVE FLANGE	1	EACH
P06556400	SHIM	3	EACH
U80045704	EXCLUDER	2	EACH
U80071214	ROTARY VARISEAL	8	EACH
U80140403	DOWEL PIN	2	EACH
U80163100	O-RING	5	EACH
U80163101	O-RING	8	EACH
U80163102	O-RING	5	EACH
U80163103	O-RING	3	EACH
U80163104	O-RING	3	EACH
U80163105	O-RING	8	EACH
U80163106	O-RING	3	EACH
U80163200	BACKUP RING	12	EACH
U80163201	BACKUP RING	6	EACH
U80163202	BACKUP RING	6	EACH
U80163203	BACKUP RING	18	EACH
U80163204	BACKUP RING	6	EACH
U80163205	BACKUP RING	18	EACH
U80163206	BACKUP RING	12	EACH
U80166500	SLOTTED SCREW	2	EACH
U80181400	O-RING	5	EACH
U80200902	SEAL	5	EACH
U80200903	SEAL	5	EACH
P04901500	SETSCREW ASSY	16	EACH
P04902300	CAP SCREW	8	EACH
P04903000	SETSCREW ASSY	4	EACH
P05036800	SHIM SET	1	EACH
U80029511	BALL BEARING	1	EACH
U80034600	NEEDLE ROLLER BEARING	1	EACH
U80035401	NEEDLE ROLLER BEARING	3	EACH
P04900600	SEAL SUPPORT RING	2	EACH
P04901400	SPRING ABUTMENT	2	EACH
P04902800	DOWEL PIN	3	EACH
P05837500	PLAIN BEARING	1	EACH
U80035411	NEEDLE BEARING WASHER	3	EACH
U80067308	INTERNAL CIRCLIP	1	EACH

HYDRAULIC TRAINING MOTOR MATERIAL

OEM PART NUMBER	ITEM DESCRIPTION	QTY	UNIT
4597B60H01	THRUST WASHER	1	EACH
U80067309	INTERNAL CIRCLIP	3	EACH
U80067310	INTERNAL CIRCLIP	2	EACH
U80067400	CIRCLIP	1	EACH
U80162100	EXTERNAL CIRCLIP	1	EACH
P06555600	DRIVING RING	2	EACH
P04901600	PLAIN WASHER	8	EACH
U80134401	DOWEL PIN	1	EACH
U80035412	WASHER	1	EACH
P05023000S01	THRUST COLLAR MATERIAL	1	EACH
U80051333	SPRING	2	EACH
U80056003	3/8 SPHERICAL BALL	3	EACH
U80056002	SPHERICAL BALL	2	EACH
P05026400	SLEEVE	2	EACH

MISCELLANEOUS MATERIAL

OEM PART NUMBER	ITEM DESCRIPTION	QTY	UNIT
9394C43H01	RDM Refurb Name Plate	1	EACH
9392C39H01	STEEL BAR 1.0+.1 X 4 LG	3	EACH
4542B10H01	Self Locking Screw	4	EACH
908A636H11	LOCTITE 85	2	EACH
922A676H02	LUBRICANT FEL-PRO C5-A	12	EACH

ROTARY JOINT

OEM PART NUMBER	ITEM DESCRIPTION	QTY	UNIT
4397B26H04	O-RING - .984" I.D.	6	EACH
4397B26H05	O-RING - 1.484" I.D.	4	EACH
4397B26H06	O-RING - 3.100" I.D.	10	EACH
8575C12H04	BACK-UP RING	16	EACH
8575C12H05	BACK-UP RING	10	EACH
8575C12H06	BACK-UP RING	24	EACH
8575C37H02	FLANGE	1	EACH